

Electronic Companion

Neural Differential Utility: Accepted Adaptive Service Control and Certified Value-Gradient Learning

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This companion contains current acceptance, accounting, switching, and computational details. Every predecessor main-paper section moved here remains intact. The separate historical supplement retains the full earlier diffusion and continuous-time development and all earlier computational tables; cross-document references identify their location. No historical numerical evidence is pooled with the nonlinear multistage experiment.

EC.1 Accepted contracts and the value of information

The relevant static comparator is chosen before period zero, not fixed at an arbitrary tier. Let $\mathcal{P}_{\text{stat}}$ consist of constant-contract protocols, with stock still optimized using the observed regime. Every comparator pays the initial installation cost $\lambda(\theta - q_0)^2$. Let $\mathcal{P}_t$, $\mathcal{P}_{t,z}$, $\mathcal{P}_{t,q}$, and $\mathcal{P}_{t,z,q}$ denote deterministic contract rules using the displayed information. Physical stock can use the full observed state in each class. With a fixed initial contract,

$$\mathcal{P}_{\text{stat}} \subseteq \mathcal{P}_t \subseteq \mathcal{P}_{t,z} \subseteq \mathcal{P}_{t,z,q}. \quad (\text{EC.1})$$

The inherited-only class is not an additional information increment along this chain.

Proposition EC.1.1 (Inherited-only information and optimized static contracts). *For deterministic q_0 and deterministic protocols, the value of $\mathcal{P}_{t,q}$ equals the value of $\mathcal{P}_t$. For a finite stock menu and continuous tier, the optimal static contract can be found exactly by enumerating stock-envelope breakpoints and the stationary point of the objective on each intervening interval.*

Proof. If $\theta_t = f_t(q_t)$ and $q_{t+1} = \theta_t$, induction makes the entire tier path deterministic. A time-only schedule implements that path, and every schedule is an inherited-only rule. Independent private randomization is a mixture of deterministic paths and cannot improve the unconstrained maximum.

For the static calculation define $a_z = \sum_t \beta^t \Pr(z_t = z)$ and $H = \sum_z a_z$. Its objective is

$$W_{\text{stat}}(\theta) = \sum_z a_z \{r_z \theta + U_z(\theta)\} - mH\theta^2 - \lambda(\theta - q_0)^2. \quad (\text{EC.2})$$

Each U_z is a finite upper envelope of affine stock payoffs. Between all pairwise line crossings, the active stock choices are constant, so (EC.2) is a concave quadratic with coefficient $-(mH + \lambda)$. Its maximum on that interval is at its stationary point or an endpoint. Checking all such candidates and the endpoints 0, 1 is exhaustive, including ties and nonconcavity across intervals. One may instead first construct each stock upper envelope and enumerate only its surviving breakpoints; intersections never attaining that envelope are unnecessary. Both procedures return the same exhaustive candidate set of active intervals. $\square$

Public randomized accepted classes are treated separately in Proposition 5.2; the unconstrained mixture argument above is not sufficient for them. The equality in the proposition concerns the value from the specified initial state; it is not equality of the two feedback function spaces on unreachable histories. For $\mathcal{P}_{t,z}$ we use occupation variables and binary tier selectors shared across all inherited states. Optimizing an ordinary full-state Bellman equation and then deleting q from the policy label would not solve this restriction.

EC.1.1 A constrained operating protocol

The customer-accepted problem is

$$\max_{\pi} W^{\pi} \quad \text{subject to} \quad F^{\pi} \geq \gamma D, \quad U^{\pi} \geq \underline{U}, \quad C^{\pi} \leq \overline{C}, \quad (\text{EC.3})$$

where the optional final constraint rules out financing provider gains with a higher physical resource cost. For a finite tier menu let $x_t(z, q, S, \theta)$ be the undiscounted state–action occupancy. It satisfies $x \geq 0$ and

$$\sum_{S, \theta} x_0(z, q, S, \theta) = \mu_0(z, q), \quad \sum_{S, \theta} x_t(z, q, S, \theta) = \sum_{z-, q-, S-} P_{z-z} x_{t-1}(z-, q-, S-, q), \quad t \geq 1. \quad (\text{EC.4})$$

All objectives and commitments are linear functionals of x , with β^t in their coefficients. The transition on the right has next contract q , explaining the last action index. The full stock menu is retained: eliminating stock by the unconstrained newsvendor solution is generally invalid under service and participation constraints.

Theorem EC.1.2 (Acceptance, Pareto comparison, and a certificate). *For finite states and actions, (EC.3) equals the linear program defined by (EC.4) and its linear commitments. From any feasible occupation measure, choosing actions with probabilities proportional to the local occupancies implements the same reward and commitments. If a static comparator is feasible, the optimum in a policy class containing it is at least its reward. Moreover, for any $\mu, \eta, \chi \geq 0$,*

$$W_{\text{acc}}^* \leq V_0^{\mu, \eta, \chi} - \mu \gamma D - \eta \underline{U} + \chi \overline{C}, \quad (\text{EC.5})$$

where $V^{\mu, \eta, \chi}$ is the ordinary Bellman value for reward $g + \mu f + \eta u - \chi C$. A feasible protocol and this upper bound give a two-sided certificate without trusting an optimizer’s status label.

Proof. History-dependent policies induce occupancies satisfying the flow equations. Conversely, normalize each positive local occupancy; induction on t recovers every occupancy, choosing arbitrary actions at zero-mass states. This proves equality and implementability. The finite feasible flow polytope is compact, and linear programming duality applies when the commitment set is nonempty. The comparison follows by feasibility. For the certificate, each feasible policy satisfies $W^{\pi} \leq W^{\pi} + \mu(F^{\pi} - \gamma D) + \eta(U^{\pi} - \underline{U}) - \chi(C^{\pi} - \overline{C})$. Maximize the expression over all policies and evaluate its reward term by Bellman recursion. $\square$

The same Lagrangian retains the inherited-contract envelope because the added commitments depend on current stock and tier, not separately on inherited q . However, the premium–liability monotonicity assumptions may change after reweighting. Neither a constrained optimizer nor its randomized policy is assumed to inherit the relaxation’s statewise monotonicity without checking those conditions.

For the canonical primitives in Section H.12, optimize the continuous static tier and set $(\gamma D, \underline{U}, \bar{C})$ equal to its fill, customer utility, and physical cost. The exact stationary candidate is

On the 0.1 tier grid, a dynamic protocol achieves

with exactly equal F , U , and C to the continuous static comparator. The finer static class makes this comparison more demanding than comparison with the best static grid point. Randomizing over static contracts cannot improve their unconstrained maximum; the static optimum itself satisfies these commitments.

This statement has a constructive rational certificate. Every positive-occupancy stock is $S = z + 2$. The protocol is deterministic except at $(t, z, q) = (1, 2, 0.5)$, where it selects tier 0.8 with probability

and tier 0.7 otherwise. The complete feedback map is supplied with the certificate. Fixing that single choice low or high gives two deterministic protocols with rational rewards and utilities. Set $\omega = (\underline{U} - U_L)/(U_H - U_L)$; exact forward propagation verifies $0 \leq \omega \leq 1$ and equality of all three commitments. The gain in (EC.6) is the positive rational number

An independent rational Bellman recursion for (EC.5), using nonnegative rational prices and $\chi = 0$, gives an upper bound less than 8.21×10^{-16} above this feasible reward. Thus the observed improvement is neither Monte Carlo noise nor a payment for lower service. It is an instance-specific, reproducible Pareto improvement in provider reward at the customer's reservation utility and service level; it is not a universal dominance theorem for every set of primitives.

For a policy π , let C^π be its expected discounted physical cost and let K^π be its expected discounted premium minus liability, maintenance, and adjustment. Its net reward is $W^\pi = K^\pi - C^\pi$. All these quantities use the same external coefficients. Let J denote a net-reward-optimal joint policy, F a frozen-contract policy optimized for that reward, and O a physical-cost-optimal policy with the same information and admissible stock actions.

Theorem EC.2.1 (Accounting and an optimizer-based physical-cost bound). *For these actual policies,*

$$C^F - C^J = (W^J - W^F) - (K^J - K^F), \quad (\text{EC.7})$$

$$0 \leq C^J - C^O \leq K^J - K^O. \quad (\text{EC.8})$$

Thus J improves physical cost over F exactly when its optimized net-reward gain exceeds the increase in net contract transfers. Physical-cost dominance over O cannot follow from net-reward optimality. A physical oracle may have multiple contract implementations; (EC.8) holds for any specified feasible implementation.

Proof. Subtract $W = K - C$ for the two policies to obtain (EC.7). Physical optimality gives $C^O \leq C^J$. Net-reward optimality gives $K^J - C^J \geq K^O - C^O$, yielding the other inequality. Neither step inserts a preferred contract into the optimizing policy. $\square$

The bound can be strengthened by choosing the net-transfer-maximizing implementation within the physical-optimal policy set. Our primary experiment does not change the previously recorded oracle implementation for this purpose: the greatest-contract tie convention selects contract one. The physical lower bound is independent of that convention, whereas the oracle's reported net cash flow is not.

A positive physical gap can also be stable without assuming that all reoptimized policies remain unchanged. Suppose the same two calibrated policies are deployed under a perturbed exogenous one-step law whose total-variation distance is at most η uniformly in regime, and physical stage cost lies in $[0, \bar{C}]$. Coupling exogenous histories gives

$$|(C^F - C^J)_{\text{new}} - (C^F - C^J)_{\text{old}}| \leq 2\bar{C}\eta \sum_{t=0}^{T-1} \beta^t (t+1). \quad (\text{EC.9})$$

Before a coupling failure the two copies of each fixed policy choose identical contracts and stock. The chance of failure by period t is at most $(t+1)\eta$; summation proves the bound. This statement concerns deployment of the calibrated policies, not arbitrary policy reselection at a tie. The traditional critical-fractile mismatch argument is retained separately in EC.2; it is not used to attribute physical improvement to an unproved endogenous selector.

EC.2.1 The actual multi-tier inaction regions

A pairwise crossing alone need not be a surviving decision region. The following statement compares every competing tier. Fix the continuation value and let a one-review premium shift s add $s\theta$ to its intrinsic reward B_i ; future premiums are not changed. For inherited tier θ_j , define

$$\ell_j = \max_{i < j} \left\{ \frac{B_i - B_j}{\theta_j - \theta_i} - \lambda(\theta_j - \theta_i) \right\},$$

$$u_j = \min_{i > j} \left\{ \frac{B_j - B_i}{\theta_i - \theta_j} + \lambda(\theta_i - \theta_j) \right\},$$

with the empty maximum $-\infty$ and empty minimum $+\infty$.

Proposition EC.2.2 (Surviving inaction bands). *Under the greatest-tier tie convention, the installed tier θ_j is retained exactly when $\ell_j \leq s < u_j$. An empty interval means that this tier has no inaction band. For a uniform menu of width h and concave intrinsic reward sequence, the finite endpoints reduce to*

$$\ell_j = -\frac{B_j - B_{j-1}}{h} - \lambda h, \quad u_j = -\frac{B_{j+1} - B_j}{h} + \lambda h.$$

Both endpoint sequences are nondecreasing. Each interior band's width is $[(B_j - B_{j-1}) - (B_{j+1} - B_j)]/h + 2\lambda h$, and adjacent bands overlap by exactly $2\lambda h$.

Proof. Subtract the payoff from every lower and higher competing tier at inherited state $q = \theta_j$. Every lower comparison imposes the corresponding lower bound, whereas a tie with a higher tier excludes the upper endpoint. Their intersection is precisely the displayed interval. Under concavity, the largest lower restriction and smallest upper restriction come from the adjacent tiers. Ordered adjacent secant slopes give ordered endpoints and the stated width and overlap formulas. $\square$

Concavity is a sufficient condition for the simplified formulas, not an implication of the general inventory envelope. Our diagnostic therefore uses all competitors for each of the 264 period–regime–tier cases and stores empty bands as well as surviving ones. A choice cell in the inherited coordinate is not called an inaction interval when the chosen tier differs from that coordinate. Neither this result nor a fixed-continuation shift is used as a permanent-premium comparative-static theorem.

EC.3 Continuation participation and its rational certificate

Index the regime-history decision nodes by i , with discounted probability $w_i > 0$, parent $p(i)$, and tier x_i . For the root, the parent tier is the fixed q_0 . Let a_i denote its net payment rate. The interim constraints can be multiplied by the unconditional discounted probability of their root without changing their meaning. Their matrix form is

$$Cx \leq h, \quad C_{ij} = w_j a_j \mathbf{1}_{\{j \text{ descends from } i\}}, \quad h_i = \bar{\theta} \sum_{j \text{ descends from } i} w_j a_j.$$

Write the reward as $W(x) = -x^\top Hx + b^\top x - c_0$. The adjustment graph contributes nonnegative squares, so $H \succeq m \text{diag}(w)$. The change of variables $y_i = \sqrt{w_i} x_i$ gives a well-conditioned sparse quadratic matrix for the numerical proposal; a sparse factorization is reused in the bound-constrained dual solve. There are only $O(TN)$ nonzero continuation coefficients on an N -node finite tree, rather than a dense N^2 constraint matrix.

Proposition EC.3.1 (Weighted residual certificate). *For any feasible $x \in [0, 1]^N$, nonnegative vector ℓ , and residual $r = b - 2Hx - C^\top \ell$,*

$$W^{\text{interim},*} - W(x) \leq \ell^\top (h - Cx) + \sum_i \frac{r_i^2}{4mw_i}.$$

All terms can be computed exactly from rational node probabilities, tier choices, and multipliers, even when a floating-point optimizer proposed them.

Proof. The Lagrangian $W(y) + \ell^\top (h - Cy)$ is an upper bound on every feasible reward. Its difference from its value at x is $r^\top (y - x) - (y - x)^\top H(y - x)$. Bound H below by $m \text{diag}(w)$ and complete each scalar square. Adding the slack term proves the result. Optimizing over all real vectors only relaxes the tier box. $\square$

The numerical proposal is rounded down to denominator 10^9 . If a continuation budget is violated after rounding, multiply all tiers by the smallest ratio of the corresponding budget to its proposed subtree payment, then round down again. Since every a_i is positive, this enforces every subtree constraint and preserves the tier box. In the recorded calculation the contraction is approximately 0.99999943, not a material change in the policy. The proposed nonnegative dual variables are independently rounded; no stationarity claim is inferred from solver success.

The resulting deterministic reward is -5.4170322098230885 , and the exact upper-minus-lower gap is $6.948160845151524 \times 10^{-7}$. The tier range is approximately $[0.087564039, 0.672234204]$. All 3,280 continuation inequalities, the customer root constraint, physical cost, and service are satisfied. The displayed gap includes both weighted stationarity error and complementary slackness, and does not drop the larger of the two. The JSON certificate contains every rational tier and multiplier in breadth-first tree order. A separate verifier reconstructs the tree and repeats the audit without the numerical optimizer.

EC.4 Expanded actions and the provider relaxation

The following predecessor development is retained without changes to its theorem statements or proofs. Its unconstrained value is not an accepted value.

EC.4.1 Exact representation and comparator classes

Proposition EC.4.1 (Expanded actions and the fixed-contract restriction). *The model in (5) is exactly the Markov decision process with state (z, q) , action (S, θ) , reward (2), and transition $P_{zz'} \mathbf{1}_{\{q'=\theta\}}$. Its value is unchanged by flattening the product action set into one action index. A frozen-contract problem is the restriction $\theta = \bar{\theta}$; its value cannot exceed the joint value when the same initial state and external reward are used. A dummy label that changes neither reward nor the next state does not enlarge the attainable value.*

Proof. A policy in either representation assigns the same pair to each observed history, so it induces the same joint law of states, actions, and rewards. Equivalently, the two Bellman maxima enumerate the same quantities. The frozen problem optimizes over a subset of these policies. Duplicating a maximand leaves a maximum unchanged. $\square$

Optimizing over a genuine extra decision can nevertheless produce a reduced operator that is not affine in the continuation value. For fixed physical action, write

$$\mathcal{B}(v) = \sup_{\theta} \{b_{\theta} + \beta \langle P_{\theta}, v \rangle\}. \quad (\text{EC.10})$$

If this envelope has two different uniquely exposed slopes on an open convex set of continuation values, it cannot equal one fixed affine reward/kernel primitive there. This is a statement about eliminating a decision while retaining one primitive, not a separation from the expanded-action problem. The proof and the precise exposed-slope condition appear in Electronic Companion (EC) Section EC.1.

Nor does support-envelope geometry exclude entropic control. For $\rho > 0$, a finite full-support reference distribution P , and any bounded vector v ,

$$\frac{1}{\rho} \log \mathbb{E}_P e^{\rho \beta v} = \sup_{Q \in \Delta} \left\{ \beta \mathbb{E}_Q v - \frac{1}{\rho} \text{KL}(Q \| P) \right\}. \quad (\text{EC.11})$$

The optimizer is proportional to $P \exp(\rho \beta v)$. Subtracting the objective at a particular Q from the left side gives $\rho^{-1} \text{KL}(Q \| Q^*) \geq 0$. A sufficiently rich kernel-control family therefore represents this entropic operator exactly. The inventory model uses its specific contractual family; the general framework does not claim non-equivalence to all risk-sensitive or robust-control models.

EC.5 Structure of the provider relaxation

Throughout this section, $V = V^{\text{rel}}$ denotes the provider relaxation. Accepted values are denoted W^{acc} or V^{acc} ; a relaxation property is not transferred to an accepted optimizer without checking its constraints.

Assumption EC.5.1 (Premium–liability compatibility). Demand is stochastically nondecreasing in z , the regime transition is stochastically monotone, and for every $z_2 \geq z_1$ and feasible S ,

$$r_{z_2} - r_{z_1} \geq \kappa \{L_{z_2}(S) - L_{z_1}(S)\}. \quad (\text{EC.12})$$

All other cost coefficients and the product action set are independent of z .

For a translated demand family $D_z = D_0 + z$, shortage is 1-Lipschitz in the translation. Thus $r_{z_2} - r_{z_1} \geq \kappa(z_2 - z_1)$ is a sufficient primitive condition. It allows shortage exposure to rise; premium growth must compensate for the incremental liability attached to a larger contract.

Theorem EC.5.2 (Monotone stock and contract). *Under Assumption EC.5.1, V_t is supermodular in (z, q) for every t . The set of maximizing pairs in (5) is a compact sublattice and has a greatest element. Both components of this greatest optimizer, $S_t^*(z, q)$ and $\theta_t^*(z, q)$, are nondecreasing in z and in q .*

Proof. We verify increasing differences for the full maximand, including the continuation value. In the action pair (S, θ) , the only cross term is $-\kappa \theta L_z(S)$, which has increasing differences because L_z decreases in S . In (θ, q) , the cross difference from adjustment is $2\lambda \Delta \theta \Delta q \geq 0$.

For fixed θ , $-C_z(S) - \kappa \theta L_z(S)$ has increasing differences in (S, z) . To see this without a density assumption, its stock marginal at continuity points is

$$-c - hF_z(S) + (p + \kappa \theta)(1 - F_z(S));$$

stochastic demand ordering makes this marginal nondecreasing in z . Integrating between two stock levels proves the required finite-difference inequality, including at atoms. The stage cross difference in (θ, z) is

$$\Delta\theta [\Delta r - \kappa\Delta L(S)] \geq 0$$

by (EC.12). All remaining cross pairs involving q or z have zero stage cross difference.

Suppose V_{t+1} is supermodular in (z', q') . For $\theta_2 \geq \theta_1$, the function $z' \mapsto V_{t+1}(z', \theta_2) - V_{t+1}(z', \theta_1)$ is nondecreasing. Stochastic monotonicity here means $\sum_{z'} P_{z_2 z'} f(z') \geq \sum_{z'} P_{z_1 z'} f(z')$ for every increasing f and $z_2 \geq z_1$, equivalently ordering every upper-tail probability. It therefore makes its conditional expectation nondecreasing in z . The continuation term has increasing differences in (z, θ) and is independent of S, q . Consequently the full maximand is supermodular on the product lattice (S, θ, z, q) .

Maximizing a supermodular function over a fixed compact action lattice preserves supermodularity in the remaining coordinates: apply the lattice inequality to maximizing pairs at two parameter states, and use their meet and join as feasible pairs at the meet and join of those states. The same inequality makes the argmax a sublattice and orders its greatest elements when the parameters increase. To make the selection step explicit, take greatest maximizers a at x and b at $y \geq x$. The joint lattice inequality gives $F(a, x) + F(b, y) \leq F(a \wedge b, x) + F(a \vee b, y)$. Optimality forces equality, so $a \vee b$ is optimal at y ; greatest selection implies $a \vee b \leq b$, hence $a \leq b$. Continuity and compactness give nonempty compact argmax sets. Closure under meet and join gives their greatest elements on the finite-dimensional compact product of chains. This is the supermodular-maximization and greatest-selection form of the lattice theorem of Topkis (1978), with its existence and order hypotheses verified here. Since $V_T = 0$ is supermodular, backward induction completes the proof. $\square$

The result concerns the state variables of the stated model. It does not assert a backlog comparative static in a model that has no carried backlog. It also does not follow for an internal penalty-only objective. With terminal value zero, $p(\theta) = \theta$, $q = 1/2$, and fixed stock, that alternative objective has optimizer

$$\arg \max_{0 \leq \theta \leq 1} \{-\theta L - \lambda(\theta - 1/2)^2\} = \text{proj}_{[0,1]} \left(\frac{1}{2} - \frac{L}{2\lambda} \right), \quad (\text{EC.13})$$

which decreases with exposure. External premium income in (2), and specifically (EC.12), changes this conclusion.

EC.5.1 Stock elimination and the inherited-contract envelope

For a fixed contract, stock solves a newsvendor problem:

$$S_z(\theta) \in \arg \min_{S \in \mathcal{S}} \{cS + hO_z(S) + (p + \kappa\theta)L_z(S)\}. \quad (\text{EC.14})$$

On a stock interval with a continuous strictly increasing demand distribution, an interior optimum has critical fractile

$$F_z(S_z(\theta)) = \frac{p + \kappa\theta - c}{h + p + \kappa\theta}. \quad (\text{EC.15})$$

Boundary solutions apply when the right side is outside the attainable distribution range. For atoms or a stock grid, (EC.14) itself, with the greatest-minimizer convention, is the definition; no continuous quantile formula is silently substituted for grid optimization.

Set $U_z(\theta) = \max_S \{-C_z(S) - \kappa\theta L_z(S)\}$ and

$$b_t(z, \theta) = r_z\theta - (m + \lambda)\theta^2 + U_z(\theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta). \quad (\text{EC.16})$$

Then

$$V_t(z, q) = -\lambda q^2 + \max_{\theta \in \Theta} \{b_t(z, \theta) + 2\lambda\theta q\}. \quad (\text{EC.17})$$

Theorem EC.5.3 (Exact envelope reduction). *For any demand family and any regime transition, $q \mapsto V_t(z, q) + \lambda q^2$ is convex. If Θ contains N_θ ordered points, this function is a convex piecewise-affine envelope with ordered slopes $2\lambda\theta$. Its active line selects a greatest optimal contract, and (EC.14) supplies the corresponding stock. After stock costs and continuation expectations are computed, the envelope can be constructed in $O(N_\theta)$ arithmetic operations from the ordered slopes and queried at N_q ordered inherited-contract states in $O(N_\theta + N_q)$ operations. No global concavity of b_t is required.*

Proof. Equation (EC.17) is obtained by expanding $-\lambda(\theta - q)^2$. A supremum of affine functions is convex. For finite Θ , insert lines in increasing slope order. Intersections with the current last line must increase along an upper hull; otherwise the last line has an empty interval of optimality and is removed. Each line is inserted and removed at most once. At an intersection choose the larger slope, which selects the greater contract. A single forward pass through ordered queries locates all active intervals. Stock elimination is exact because the continuation and adjustment terms do not depend on stock. $\square$

The operation count is an arithmetic count, not a bit-complexity statement for arbitrarily long rational numbers. A direct factored enumeration already reduces the expensive product search: on the recorded grid, it uses 2,904 rather than 37,752 Bellman action comparisons, plus 429 one-time stock-stage computations. The hull construction further avoids enumerating every contract at every inherited state. The executed hull and factored enumeration agree with exhaustive rational optimization at all 264 nonterminal states.

EC.5.2 A quadratic bound without global concavity

Let V now allow every contract in $[0, 1]$, retaining the same stock set. Let $V^{(n)}$ restrict all contract actions to $\Theta_n = \{0, 1/n, \dots, 1\}$ but permit any initial $q \in [0, 1]$. After the first action, the inherited state lies on the grid. Define $H_r = \sum_{k=0}^{r-1} \beta^k$, with $H_0 = 0$.

Lemma EC.5.4 (A nonsmooth global-maximizer lemma). *Let G be finite, convex, and continuous on $[0, 1]$, let $A \geq 0$, and set $f(x) = G(x) - Ax^2 + \ell x$. If an interior point x^* is a global maximizer, every $g \in \partial G(x^*)$ equals $2Ax^* - \ell$, and $f(x^*) - f(y) \leq A(y - x^*)^2$ for every $y \in [0, 1]$. A global maximum at an endpoint is represented exactly on every endpoint-containing grid. Consequently an n -interval uniform grid loses at most $A/(4n^2)$.*

Proof. A supporting line gives $f(x^* + h) \geq f(x^*) + (g - 2Ax^* + \ell)h - Ah^2$. If the linear coefficient were nonzero, a sufficiently small interior h of its sign would strictly improve the objective. Thus it vanishes for every subgradient. The same supporting inequality at any y gives the stated bound, and a nearest grid point has distance at most $1/(2n)$. If G has a genuine kink, its subdifferential is not a singleton and that point cannot be an interior maximum of this form. No endpoint subgradient is assumed and no global concavity of f is used. $\square$

Theorem EC.5.5 (Continuous-contract value enclosure). *With the primitives of (2), for $r = T - t \geq 1$,*

$$0 \leq V_t(z, q) - V_t^{(n)}(z, q) \leq \frac{(m + \lambda)H_r + \beta\lambda H_{r-1}}{4n^2}. \quad (\text{EC.18})$$

This is also a performance-loss bound for the exact grid-optimal policy when it is implemented in the continuous-contract problem from (z, q) . It is uniform in z, q and does not require differentiability of demand distributions, global concavity, or a numerical smoothness estimate.

Proof. First consider $f(x) = G(x) - Ax^2 + \ell x$ on $[0, 1]$, where G is finite and convex and $A > 0$. If a maximum x^* lies at an endpoint, it belongs to every grid. If it is interior, take any subgradient s of G there. The supporting-line inequality gives

$$f(x) \geq f(x^*) + (s - 2Ax^* + \ell)(x - x^*) - A(x - x^*)^2.$$

An interior maximum forces the linear coefficient to vanish: otherwise a sufficiently small displacement with its sign would improve the objective. Therefore the nearest grid point loses at most $A/(4n^2)$.

For the Bellman maximization after stock elimination, U_z is convex as a supremum of affine functions of θ . Theorem EC.5.3, applied to continuous actions, says that $V_{t+1}(z', \theta) + \lambda\theta^2$ is convex. Consequently its contract objective has the displayed form with $A = m + \lambda + \beta\lambda$ whenever $t < T - 1$. At the final period, $A = m + \lambda$ because $V_T = 0$. Restricting just the current maximization thus loses at most the corresponding $A/(4n^2)$.

Let $E_t = \sup_{z, q} (V_t - V_t^{(n)})$. Monotonicity and the β -Lipschitz property of the Bellman operator give

$$E_t \leq \frac{m + \lambda + \beta\lambda \mathbf{1}_{\{t < T-1\}}}{4n^2} + \beta E_{t+1}, \quad E_T = 0.$$

Summing proves (EC.18). A grid policy uses admissible continuous actions and has exactly the grid-model reward and transition law, so its continuous-model value is $V^{(n)}$. $\square$

The theorem bounds contract discretization, not physical-stock discretization or an inventory diffusion limit. It avoids interpreting a small difference between two numerical grids as a proof. In the recorded instance, the 160-interval result encloses the continuous-contract initial value in $[-4.99278573, -4.99260152]$; the exact rational endpoints are supplied with the code.

EC.6 Compatible learning with an acceptance certificate

The accepted value function supplies both a reconfiguration rule and an audit of an approximate rule. A global sup-norm residual is not the only possible audit. On the identified quadratic model, the economically relevant loss has a direct energy representation.

Theorem EC.6.1 (Accepted policy-loss identity). *Suppose the feasible physical policy is identified as in Lemma 5.1, and the interior Bellman recursion (26) has a nonnegative multiplier η^* whose optimal protocol satisfies $P^* = \bar{P}$. For any accepted protocol π , including a learned or randomized one,*

$$W^{\text{acc},*} - W^\pi = \mathbb{E}^\pi \sum_t \beta^t K_t [\theta_t - \theta_t^{\eta^*}(z_t, q_t)]^2 + \eta^*(\bar{P} - P^\pi). \quad (\text{EC.19})$$

The identity concerns continuous tier actions and inherited states, not a scan of a finite chain. Certified upper bounds on the energy and the acceptance slack give a policy-loss certificate on that continuous-action model.

Proof. Completion of squares in each Bellman maximization gives its exact action gap as $K_t [\theta - \theta_t^{\eta^*}(z, q)]^2$. Sum along the law of π ; the discounted continuation terms telescope and the terminal value is zero. The left side before adding the commitment constant is $J_0^{\eta^*,*} - (W^\pi - \eta^* P^\pi)$. Since $W^{\text{acc},*} = J_0^{\eta^*,*} + \eta^* \bar{P}$, rearrangement gives (EC.19). No smoothness assumption on the learned policy or uniform residual bound enters the argument. $\square$

The identity does not make solving every control problem free: its coefficients must be obtained and verified, and its expectation must be bounded for the deployed policy. In the scalar affine case, first and second moments suffice. In a more general model a valid dual value or residual inequality is still needed. The next experiment uses such an inequality and does not require a tensor grid over its inherited contractual state.

EC.6.1 A coupled portfolio at a service review

Consider a customer buying d metered services under one portfolio agreement. Each service has the same physical primitives as above, with its observed regime fixed at this review. Aggregate fill and physical-cost commitments identify stock by summing (23) over services. The net payment vector a is fixed in the menu. Let $q \in [0, 1]^d$ be the installed tier vector. In addition to maintenance $m\|\theta\|^2$, the provider incurs $\zeta\theta^\top L\theta$ to maintain different configurations across linked services, where L is the graph Laplacian of the publicly specified service-dependency graph and $\zeta \geq 0$. This is a nonseparable configuration cost, not independent copies labeled as a high-dimensional problem.

At the current review, or at a terminal Bellman update, the accepted operating problem is

$$\max_{\theta \in [0,1]^d} \{a^\top \theta - C - m\|\theta\|^2 - \zeta\theta^\top L\theta - \lambda\|\theta - q\|^2\} \quad \text{subject to } a^\top \theta \leq \bar{P}. \quad (\text{EC.20})$$

Here acceptance is by the common portfolio buyer, not a claim of separate participation by every service's end user. For $q = \frac{1}{2}\mathbf{1}$, the uniform static tier is optimized continuously; $L\mathbf{1} = 0$ gives

$$\bar{\theta} = \frac{d^{-1}\mathbf{1}^\top a + \lambda}{2(m + \lambda)}, \quad \bar{P} = \bar{\theta}\mathbf{1}^\top a.$$

The experiment isolates a coupled, many-service accepted update. It is not presented as a 1,024-dimensional replication of the entire eight-review information experiment or as a generic nonquadratic inventory solution.

Put $K = (m + \lambda)I + \zeta L$ and $b_\eta = (1 - \eta)a + 2\lambda q$. The unconstrained quadratic Lagrangian critic is

$$J^\eta(q) = -C - \lambda\|q\|^2 + \frac{1}{4}b_\eta^\top K^{-1}b_\eta. \quad (\text{EC.21})$$

It is the actual operating value whenever its maximizer is interior. The training queries below satisfy that condition by the discrete maximum principle for K : its inverse is nonnegative and $K^{-1}\mathbf{1} = (m + \lambda)^{-1}\mathbf{1}$. In the audit, an unconstrained Lagrangian remains an upper bound even if a different candidate reaches the box boundary.

For a symmetric polynomial filter $p_c(K)$, define the learned scalar critic by replacing K^{-1} in (EC.21). Its derivative is compatible by construction, and its associated actor is

$$\hat{\theta} = q + \frac{\nabla_q \hat{J}^\eta(q)}{2\lambda} = \frac{1}{2}p_c(K)b_\eta. \quad (\text{EC.22})$$

The computation is a graph-filter network with fixed message-passing features and a trained scalar quadratic readout. Polynomial graph filters are established constructions (Defferrard et al., 2016); the new use here is accepted tier control with an independent deployment certificate, not a claim to have invented graph convolution or polynomial inversion.

Theorem EC.6.2 (A computable accepted deployment bound). *Let $K \succeq \mu I$ with $\mu > 0$. For any rationally specified feasible tier vector $\hat{\theta}$, any $\eta \geq 0$, and $r = (1 - \eta)a + 2\lambda q - 2K\hat{\theta}$,*

$$0 \leq W^{\text{acc},*} - W(\hat{\theta}) \leq \frac{\|r\|^2}{4\mu} + \eta(\bar{P} - a^\top \hat{\theta}). \quad (\text{EC.23})$$

Thus rational input coefficients, deployed tiers, and multiplier yield an exact arithmetic upper bound without solving the optimal program or sampling the inherited-state domain.

Proof. Weak Lagrangian duality bounds the accepted optimum by the unconstrained maximum of $b_\eta^\top \theta - \theta^\top K \theta$ plus its constants. Expanding about $\hat{\theta}$ leaves $r^\top d - d^\top K d \leq \|r\|^2/(4\mu)$. Adding the nonnegative commitment slack gives (EC.23). Maximizing over all real d only enlarges the upper bound, so the argument does not require an interior optimal box solution. $\square$

This bound is an ex post guarantee for the specified operational instance and deployed vector. It is not a distribution-free prediction of neural performance on an unseen graph. Each unseen graph is audited separately. Unlike the earlier finite-chain residual budget, the arithmetic of this audit is exact, and its scale can be compared directly with the operational improvement.

EC.6.2 Equal information, transfer, and classical baselines

We fix $m = 1.5$, $\lambda = 0.6$, and degree-12 Chebyshev features on the spectral interval $[2.1, 26.1]$. Graphs consist of a cycle and a random matching, so their maximum degree is at most three; $\zeta \in \{1, 2, 3, 4\}$ gives the stated spectral enclosure. The network has thirteen trained readout coefficients, independent of the number of services. Its feature layers require sparse matrix–vector products, not a learned dense inverse.

Each of ten training seeds supplies 96 independently generated graphs of sizes 32 or 64. For each graph, we make exactly three scalar teacher queries at $q - hu$, q , and $q + hu$, with $h = 0.025$ and a random sign vector u . We use the operational value (EC.21) at $\eta = 0.2$, after removing its known $-C - \lambda\|q\|^2$ terms. Both methods receive all 288 scalar observations. Value-only fitting minimizes their squared errors. Value-gradient fitting adds the directional derivative discrepancy obtained by differencing those same endpoint values. There are no extra teacher values or derivative-oracle calls. Values are normalized by graph size and directional differences by its square root; both objectives use the same least-squares readout solve.

Because both the teacher and the learned scalar critic are quadratic in q , the centered directional difference is exact in real arithmetic. The gradient objective is therefore also an explicit finite-difference regularization of the same three value observations. We do not count that algebraically identical formulation as another independent baseline. We test $h = 0.01, 0.05, 0.1$ separately. Sampling seeds quantify variation in the training data; there is no nonconvex optimizer seed to conceal behind this convex readout fit. The scalar construction, the previously compiled cubic-ReLU envelopes, and the historical trained tanh network are distinct implementations.

We deploy without retraining on twelve held-out graphs: four each at 64, 256, and 1,024 services. For each graph, two filtered right-hand sides determine the multiplier that meets the payment budget. Tiers and the multiplier are rounded to denominator 10^{10} ; a one-sided correction enforces the customer budget exactly. All coordinates, every commitment, and (EC.23) are then checked using integer and rational arithmetic. A failed certificate would remain a failed deployment, not a discarded seed.

Classical comparisons include the analytic degree-12 Chebyshev inverse, dense Cholesky solution, sparse direct solution, and conjugate gradients with relative residual tolerance 10^{-4} . All solve the same accepted problem, including multiplier calibration. The graph representation is common input to all methods. Training time is reported separately; execution times do not include graph construction or the independent rational audit. Consequently a faster actor is not described as a faster end-to-end proof of optimality.

On the 1,024-service cases, the learned value-gradient actor has a median deployment time of approximately 1.21 milliseconds, versus 7.62 milliseconds for sparse direct solution and 18.01 milliseconds for dense Cholesky in the recorded environment. Conjugate gradients take approximately 1.52 milliseconds and are more accurate. The largest exact loss bound for the learned actor is 8.05×10^{-7} per service, whereas its smallest guaranteed gain over the static alternative exceeds 0.01606 per service. These are decision-useful enclosures, not bounds larger than the economic comparison. The same thirteen coefficients transfer to new graph sizes and graph structures.

The derivative-weighted fit lowers the mean realized loss modestly relative to the equal-information value-only fit, but improves six of ten paired training seeds at each test size, not every seed. The seed-level uncertainty is reported in the companion. The operational conclusion is not universal superiority of

Table 1: Accepted coupled portfolios: 1,024 persistent tier coordinates

Method	Mean loss/service	Max. bound/service	Time (ms)
Learned value–gradient	1.52e-08	8.04e-07	1.214
Learned value only	1.68e-08	9.01e-07	1.222
Analytic Chebyshev	2.69e-08	3.67e-07	1.220
Conjugate gradient	6.11e-10	3.46e-09	1.524
Sparse direct	0.00e+00	7.80e-12	7.619
Dense Cholesky	0.00e+00	7.80e-12	18.010

All deployed policies satisfy customer acceptance exactly. Learned rows aggregate ten predeclared training seeds on four fixed test graphs; other rows use those same four graphs. Loss is against the sparse-direct value. Bounds certify the rounded actions in rational arithmetic. Time is the median recorded two-right-hand-side solve plus acceptance calibration, excluding offline training and rational audit. Small direct-method loss is rounded to zero, not asserted identically zero against the unknown real-arithmetic optimum.

gradient weighting: it is that a compact learned compatible critic can deliver accepted decisions with a small, independently proved loss at a nontrivial coupled scale. The conjugate-gradient and analytic-filter results remain visible so that the contribution is evaluated against numerical linear algebra, not only against a deliberately weak neural baseline.

Scope of the retained quadratic timings. The original factorizations were not cached and omitted the independent audit. The cached re-audit in Table 8 is the relevant repeated-solve comparison; the retained table does not establish a matched-accuracy learning advantage.

EC.7 Certified learning for nonlinear multistage service control

The quadratic portfolio study tests an exactly analyzable deployment primitive. A multistage contract requires more: the action changes inherited tiers, all future reviews enter the objective, and participation must hold on every continuation subtree. We now study this full finite-tree problem with nonquadratic resources. The experiment isolates approximation from verification; it does not assume that learning dominates a classical optimizer.

EC.7.1 A common nonlinear objective and a cumulative certificate

Let $D = \text{diag}(w)$ repeat each node’s discounted probability for every service. On a binary scenario tree, let $x_n \in [0, 1]^d$, $q_0 = \mathbf{1}/2$, and set

$$\begin{aligned}
 F(x) = & \sum_n w_n \{m \|x_n\|^2 + \gamma \sum_j x_{nj}^4 + \lambda \|x_n - x_{\text{pa}(n)}\|^2\} \\
 & + \zeta \sum_n w_n \sum_{(i,j) \in E} \omega_{ij} (x_{ni} - x_{nj})^2, \quad J(p) = \max_{0 \leq x \leq 1, Ax \leq b} \{p^\top Dx - F(x)\}. \quad (\text{EC.24})
 \end{aligned}$$

Here $m > 0$, $\gamma > 0$, and graph weights are nonnegative. Every row of A gives total discounted tariff payment in one subtree; its cap equals the corresponding payment of the static outside protocol. Therefore every observed history has a participation constraint. Spatial graph edges and temporal parent–child edges appear simultaneously. All scenarios and parameters are specified before operations, while a tree action depends only on its realized history; no future realization is revealed to the controller.

The net tariff is a , and $p = a + r$ collects a linear provider resource saving r . Adding a fixed resource offset of ten per service–review makes the simulated resource costs nonnegative without changing any decision or reward difference. This extension does not represent endogenous pricing. In each central test instance, weighted-centered rational shocks set $p_i = m + \gamma/2 + e_i$ with $\sum_i w_i e_i = 0$. Hence the derivative of the constant-tier objective at one half is exactly zero. Strict concavity makes that tier the unique reoptimized static optimum, including initial installation; it is feasible for every continuation constraint. The comparison does not select a weak baseline to favor the learner.

Theorem EC.7.1 (Full-horizon nonlinear residual certificate). *For any feasible x in (EC.24) and any $\eta \geq 0$, define*

$$r = Dp - \nabla F(x) - A^\top \eta, \quad \varepsilon(x, \eta) = \frac{1}{4m} \sum_i \frac{r_i^2}{w_i} + \eta^\top (b - Ax).$$

Then $0 \leq J(p) - [p^\top Dx - F(x)] \leq \varepsilon(x, \eta)$. This is a cumulative discounted policy-value bound for the whole contingent tree, not a single-review error. With rational primitives and rational x, η , feasibility and the bound are exactly checkable, including quartic, graph, and all parent–child terms.

Proof. Since $\nabla^2 F \succeq 2mD$, for any feasible y the reward difference is at most

$$r^\top (y - x) - m\|y - x\|_D^2 + \eta^\top A(y - x) \leq r^\top (y - x) - m\|y - x\|_D^2 + \eta^\top (b - Ax).$$

Completing the square bounds the first two terms by $\|r\|_{D^{-1}}^2/(4m)$. Taking the supremum over y proves the assertion. No floating-point solver status or learned generalization premise enters this inequality. $\square$

This is the standard strong-convexity residual argument (Boyd and Vandenberghe, 2004) specialized to the continuation-constrained nonlinear tree. Its operational content is an independently replayable full-horizon upper bound. Learning supplies a candidate, not the validity of that bound.

The same upper bound supports a no-harm deployment gate. Let V_x and V_s be the exact rewards of the proposed and optimized static feasible protocols. Deploy whichever has larger reward; its upper gap is

$$V_x + \varepsilon(x, \eta) - \max\{V_x, V_s\} \geq 0. \tag{EC.25}$$

Thus failed transfer cannot force adoption of a worse protocol. For a prescribed loss tolerance, a candidate whose bound is too large must be refined by a classical solver until a valid certificate meets that tolerance; the experiments do not count such refinement as free. The present timing table measures candidate construction and audit, not a hypothetical unmeasured equal-accuracy hybrid.

For a receding-horizon implementation, the budget state in (40) must be propagated. If a feasible action and child-budget allocation incur a verified Bellman loss at most ϵ_t relative to the same continuation value, the

standard Bellman telescoping argument gives cumulative loss at most $\mathbb{E} \sum_t \beta^t \epsilon_t$. This observation requires valid continuation bounds; separate one-step neural residuals alone do not establish it. The experiment below instead certifies complete contingent plans directly by Theorem EC.7.1.

EC.7.2 Nonlinear compatible critics and equally available scalar observations

Strong convexity yields a unique $x^*(p)$ and the envelope identity $\nabla_p J(p) = Dx^*(p)$. Moreover, ∇J is Lipschitz from the D norm to its dual with constant $1/(2m)$. We fit a scalar neural potential and obtain its actor by differentiation, rather than fitting unrelated value and policy outputs. The potential uses 63 softplus features of the forcing residual and its first two local graph-tree smoothing transforms. The transforms are D -self-adjoint; the trained parameters are the readout coefficients. This finite-feature neural model is nonlinear and compatible, but is not claimed to be a new universal architecture.

Each of 24 training instances contributes the same three scalar observations at $p - hu, p, p + hu$, with $h = 1/40$ and a sign direction u . The value-only and value-gradient objectives use exactly those 72 values. Endpoint perturbations hold the tariff, outside protocol, and feasible set fixed: they are local value-function oracle queries, not new reoptimized static comparisons. Both fits use the same fixed ridge, features, and scaling. Unlike the earlier quadratic diagnostic, centered differences here are not exact derivatives. If the two endpoint value errors are bounded by ϵ_- and ϵ_+ , the directional target for the value residual, normalized by $M = \sum_i w_i$, satisfies

$$\left| \widehat{\partial}_u(J - J_s)/M - \partial_u(J - J_s)/M \right| \leq \frac{h\|u\|_D^2}{4mM} + \frac{\epsilon_- + \epsilon_+}{2hM}. \quad (\text{EC.26})$$

Indeed, integrate the directional derivative between the endpoints and use the Lipschitz constant; J_s is linear in p . All 720 scalar training records have rationally verified error bars. The largest unnormalized oracle gap is 6.143×10^{-8} in the recorded run, and the largest normalized directional error bound is approximately 0.004167. These are target-error guarantees, not a statistical guarantee for the fitted network.

A deployment first clips the actor to the box, rounds down to rational tiers, and applies a common nonnegative scaling that satisfies every subtree cap. Multipliers are proposed by nonnegative least squares and rounded; an independent verifier recomputes all derivatives and participation inequalities with fractions. The static gate then compares exact values. No failed or negative-reward proposal is discarded from the reported sample.

EC.7.3 Design and results under graph, horizon, and curvature shifts

Ten training seeds use three-review trees and four-service cycle graphs, or 28 tier coordinates. Testing uses two separately generated instances in each of four families: the training-sized cycle; an eight-service random-edge graph; an eight-service geometric graph on a four-review tree; and a weighted two-block graph with 32 services on a five-review tree. The largest contingent plan has 992 coordinates. The last two families also change quartic curvature and coupling strength. Thus topology, degree, weights, horizon, and operator spectrum can change together. This is a deliberately difficult synthetic transfer study, not evidence of calibration to a real service operation.

Classical comparisons solve exactly the same nonquadratic tree problem. Newton–direct factorizes its changing Hessian. Newton–PCG caches one baseline Hessian factorization per problem and uses it as a preconditioner for all subsequent changing Hessians and repeated solves. Setup is separated from online cost. Both methods retain the common participation dual, rational repair, and full audit. Fifteen repetitions follow one warmup; raw timing observations, setup costs, and training times are retained. Timing is not pooled with training-seed uncertainty.

Table 2 reports every outcome. All 176 deployment candidates satisfy all continuation constraints after repair, with 2,640 exact subtree checks and 52,624 exact tier-bound checks. Classical maximum normalized upper gaps are below 5.7×10^{-10} . Learned proposals create positive mean incremental reward in the cycle, random-edge, and geometric families, but both learned methods lose reward on average under the severe weighted-block shift. The static gate preserves nonnegative incremental reward without concealing those raw failures. The neural upper bounds on this shifted family are correspondingly loose; the earlier one-millionth quadratic bound is not transferred to this experiment.

Across the eight problems, the ten paired training-seed mean differences favor derivative weighting by approximately 0.000577 per discounted service–review, with a Student 95% interval of $[-0.003923, 0.005076]$. The interval includes zero. These observations do not establish a reproducible superiority claim for derivative supervision. They do establish that the accepted nonlinear multistage implementation, its adverse transfer outcomes, and its complete value certificates can be examined independently.

Table 3 includes candidate generation, repair, multiplier proposal where needed, and rational certification. Cached preconditioning materially reduces the measured cost of the largest classical case relative to repeated changing-Hessian factorization. Neural candidate times are smaller in this run, but their accuracy is different and some proposals fail the economic improvement test. Neither this table nor the quadratic portfolio table implies a repeated-solve advantage at matched certified loss.

EC.8 Current repair, uncertainty, and target-error diagnostics

This section gives the complete diagnostics supporting Section 7. The frozen design uses four training seeds (12000–12003), twelve test seeds (920000–920011) in each of four graph families, and no selection based on a favorable seed. The independent-instance sample size is 48, not the number of seed–instance–method combinations. The graph families are fixed strata. Bootstrap intervals resample seeds jointly across instances and instances within strata; they do not resample individual actor outcomes as independent observations. The complete original ten-seed/eight-instance study remains above and is not pooled with this one.

EC.8.1 Pre-repair violations, projection, and feasible actors

Table 4 distinguishes the actor before continuation repair from each repaired policy. All gains use the same objective. A common radial factor can modify every coordinate because one continuation row is violated; its cost can therefore extend beyond the node causing the violation. Weighted projection is solved by SLSQP with analytic derivatives of squared distance, followed by rational rounding and exact subtree repair. Any residual rounding correction is charged to the projection pipeline. The raw records retain both projection

Table 2: Nonlinear multistage control: mean normalized gain and exact audit. Gains divide total discounted reward by discounted service–reviews. Each learned row averages ten training seeds and two problem seeds; a classical row has two problems. The maximum certificate bound is not an estimated realized loss. “Static gate” selects the higher exact reward of the proposal and optimized static rule; no test is removed.

Family	Method	Raw gain	Static gate	Max. bound	Gate returns static
Cycle	Newton–direct	0.011549	0.011549	5.63e-10	0/2
Cycle	Newton–PCG	0.011549	0.011549	5.63e-10	0/2
Cycle	Value only	0.008528	0.008528	0.0124	0/20
Cycle	Value–gradient	0.009686	0.009686	0.00784	0/20
Random edge	Newton–direct	0.014342	0.014342	1.6e-10	0/2
Random edge	Newton–PCG	0.014342	0.014342	1.6e-10	0/2
Random edge	Value only	0.012158	0.012158	0.0221	0/20
Random edge	Value–gradient	0.013160	0.013160	0.00649	0/20
Geometric	Newton–direct	0.009155	0.009155	4.56e-10	0/2
Geometric	Newton–PCG	0.009155	0.009155	5e-10	0/2
Geometric	Value only	0.002601	0.002634	0.0397	2/20
Geometric	Value–gradient	0.003773	0.003829	0.0287	1/20
Weighted block	Newton–direct	0.001953	0.001953	5.51e-10	0/2
Weighted block	Newton–PCG	0.001953	0.001953	5.51e-10	0/2
Weighted block	Value only	-0.097068	0.000000	4.8	20/20
Weighted block	Value–gradient	-0.098095	0.000000	3.69	20/20

Table 3: Measured end-to-end candidate construction, feasibility repair, and rational audit, in milliseconds. Entries average the two problem-instance medians and their 10th–90th percentiles, each from 15 repetitions after one warmup. Learned timing uses the prespecified training seed zero, not the fastest seed. Factorization setup and training are separate. Methods do not have matched accuracy; this table is not a speedup-at-equal-quality claim.

Family	Method	Median	[10th, 90th]
Cycle	Newton–direct	14.89	[14.82, 15.00]
Cycle	Newton–PCG	22.21	[22.06, 22.44]
Cycle	Value only	6.39	[6.35, 6.55]
Cycle	Value–gradient	6.36	[6.33, 6.40]
Random edge	Newton–direct	15.09	[14.97, 15.27]
Random edge	Newton–PCG	19.21	[19.10, 19.38]
Random edge	Value only	9.87	[9.82, 9.97]
Random edge	Value–gradient	9.83	[9.79, 9.90]
Geometric	Newton–direct	39.55	[39.36, 39.75]
Geometric	Newton–PCG	48.37	[48.10, 48.69]
Geometric	Value only	17.88	[17.81, 18.05]
Geometric	Value–gradient	18.00	[17.92, 18.13]
Weighted block	Newton–direct	506.82	[505.12, 508.47]
Weighted block	Newton–PCG	269.91	[268.29, 271.84]
Weighted block	Value only	214.51	[213.73, 216.34]
Weighted block	Value–gradient	215.54	[214.32, 217.20]

solver status and all exact feasibility checks. Neither a projection status nor a neural output is used as a certificate.

The tree-transfer actor fits nonnegative flows to the predicted displacement by weighted nonnegative least squares. For a transfer on edge $u \rightarrow v$ and service j , the direction has components $1/(w_u a_{uj})$ at the parent and $-1/(w_v a_{vj})$ at the child. A positive combination has subtree payment change equal to minus the entering transfer, so all continuation caps remain satisfied. The final scalar step obeys every tier boundary. At a boundary, this step can be zero; the implementation records that outcome. A simpler segment from the comparator toward the clipped actor is also recorded, but can be identically static when every cap initially binds. Neither parameterization is claimed to span all vector-service accepted improvements, because within-node cross-service reallocations may also matter.

EC.8.2 Gate ratios and measured large-instance fallback

The denominator in Table 5 is the best recorded classical feasible gain on the corresponding original instance, not an unknown exact optimum. Its tiny certified gap is retained in the predecessor results. The proposal gap uses the updated residual calculation; the selected gap also uses a separately audited static upper bound. Both are valid upper bounds on the selected policy’s loss. The original weighted-block proposals all select static before refinement. The current fallback runs the cached Newton method afresh and audits its full contingent policy. It does not reuse a fictional learned Hessian or count the fallback as a neural success. The full timing record separates this measurement from the original cached-quadratic measurements.

EC.8.3 Timing, common setup, and endpoint errors

Table 6 uses five repetitions after one warmup on a separately identified first instance per family, with pipeline order randomized within repetition. The common outside-contract solve, sparse operator setup, and cached static certificate are listed in the raw setup file. The table includes all online construction and refinement stages. Python, NumPy, SciPy, operating-system, and thread settings are recorded with the results. The numerical stopping flag is not substituted for the common rational error threshold.

The endpoint sensitivity study regenerates twelve independent training-sized problems and solves two endpoint value queries at each of three step sizes. Tariffs, outside protocol, and participation caps are fixed during each pair. The reported finite-difference error bound is the bias term from the forcing-value gradient Lipschitz constant plus the sum of endpoint certified gaps divided by twice the step size. Additional endpoint-error allowances in Table 7 are an analytical sensitivity calculation, explicitly distinguished from newly simulated label noise. The smaller step reduces the bias bound while amplifying oracle uncertainty. A small observed mean improvement between training losses is not itself evidence that the directional targets are sufficiently accurate.

EC.8.4 Structural cross-checks and their status

The proof of the capacity theorem does not depend on numerical testing. Independent LP formulations nevertheless cross-check its directional and normal-cone forms on 180 randomly generated problems with

Table 4: Raw actors and alternative acceptance repairs

Family	Actor	Feasible before	Mean scale	Before gain	Radial gain	Projection gain	Tree-flow gain
Cycle	Value	9/48	0.933	-0.8182	-0.8228	-0.8182	0.0023
Cycle	Value–gradient	13/48	0.953	-0.7616	-0.7664	-0.7620	0.0023
Random-edge	Value	8/48	0.937	-0.8192	-0.8215	-0.8189	0.0029
Random-edge	Value–gradient	8/48	0.959	-0.7621	-0.7642	-0.7622	0.0029
Geometric	Value	9/48	0.950	-1.4773	-1.4803	-1.4765	0.0015
Geometric	Value–gradient	10/48	0.971	-1.4383	-1.4407	-1.4383	0.0018
Weighted block	Value	8/48	0.956	-2.0912	-2.0952	-2.0903	-0.0036
Weighted block	Value–gradient	9/48	0.976	-2.0669	-2.0703	-2.0668	-0.0023

Before denotes box clipping without continuation repair. Negative raw outcomes are retained. Projection minimizes weighted Euclidean actor distance, not economic loss. Tree-flow uses nonnegative child-to-parent payment transfers and a box-limited step. All reported repaired policies receive the same exact feasibility and value audit. All gains are relative to the reoptimized static protocol per discounted service–review; the comparison is not field-calibrated.

Table 5: Original nonlinear transfer: certificate size relative to available gain

Family	Actor	Mean raw gain	Proposal gap / classical gain	Selected gap / classical gain
Cycle	Value	0.008528	0.59	0.59
Cycle	Value–gradient	0.009686	0.44	0.44
Random-edge	Value	0.012158	0.43	0.43
Random-edge	Value–gradient	0.013160	0.24	0.24
Geometric	Value	0.002601	3.00	3.00
Geometric	Value–gradient	0.003773	2.31	2.31
Weighted block	Value	-0.097068	1340.86	23.07
Weighted block	Value–gradient	-0.098095	1484.75	23.07

Ratios are medians across the original two instances and ten training seeds, using the classical certified feasible gain as denominator. The selected gap uses both the proposal and static upper bounds, as in the main-paper gate equation. The weighted-block gate still has a large gap; a no-harm choice is not an accuracy certificate. All forty weighted-block pipelines were subsequently refined by the cached classical solver and independently audited.

Table 6: Timing variation separated from test-instance and training-seed variation

Family	Pipeline	Repetitions	Minimum	Median	Maximum
Cycle	Cold SLSQP	5	11.73	11.76	11.85
Cycle	Value	5	18.92	19.10	19.50
Cycle	Value–gradient	5	18.53	18.74	18.95
Random-edge	Cold SLSQP	5	41.13	41.73	42.01
Random-edge	Value	5	48.70	49.65	59.45
Random-edge	Value–gradient	5	47.67	48.55	52.80
Geometric	Cold SLSQP	5	67.37	67.63	68.11
Geometric	Value	5	91.91	93.55	93.83
Geometric	Value–gradient	5	91.43	92.89	93.77
Weighted block	Cold SLSQP	5	108.66	109.76	110.65
Weighted block	Value	5	138.82	139.56	140.86
Weighted block	Value–gradient	5	135.83	137.49	138.97

Milliseconds to the same 10^{-7} certified tolerance on the first new instance of each family, one frozen training seed, five repetitions after one warmup. Pipeline order is randomized within repetition. This separate experiment does not enlarge the economic test sample; every timing record and final policy is retained.

Table 7: Directional targets: step size and certified oracle error

Step size	Additional endpoint error allowance	Maximum measured direction error	Maximum certified target-error bound
1/80	0e+00	5.58e-07	0.00208
1/80	1e-07	5.58e-07	0.00209
1/80	1e-05	5.58e-07	0.00288
1/40	0e+00	8.89e-06	0.00417
1/40	1e-07	8.89e-06	0.00417
1/40	1e-05	8.89e-06	0.00457
1/20	0e+00	3.45e-05	0.00833
1/20	1e-07	3.45e-05	0.00834
1/20	1e-05	3.45e-05	0.00853

Twelve independently generated training-sized problems. Endpoint values are re-solved and rationally certified at every step size. The extra error column is an explicit sensitivity allowance added to each endpoint certificate, not a claim that fresh noisy labels were simulated. Measured error compares with the separately solved central actor and is numerical; the displayed bound includes oracle gaps and the finite-difference bias bound.

boundary coordinates, zero and signed commitment coefficients, several constraints, and nonconstant comparators. A separate set of 48 trees, with 3, 7, 15, and 31 nodes, compares the full balance program with the cumulative-cut critical-friction program. Thresholds are also tested just below, at, and above the returned value. These are floating-point HiGHS diagnostics, not exact theorem certificates. The two-review threshold 72/155, the nonconstant boundary optimality interval $[0, 1]$, and the first-order boundary-loss example are independently evaluated with rational arithmetic. Initial installation and outgoing switching edges are included throughout.

EC.9 Nonlinear study: reproducibility and audit details

The executable entry point is `revisions/or-r10-20260921/reproduce.sh`. The verification-only entry point is `verify.py`; it reads serialized rational inputs and does not import the optimizer or the neural feature code. Training, deployment, raw timing, structural-test, and static-gate records are separate files. Every deployment includes its full rational tier vector, nonnegative multiplier vector, exact value, exact static comparator value, and exact upper-gap fraction. Every training endpoint includes an analogous record. The former R7 accepted hierarchy, customer-exit, and quadratic portfolio records are separately regenerated and replayed; they are not relabeled as nonlinear results.

The binary tree has transition probability one half and discount factor nine tenths. Maintenance curvature is three halves, temporal adjustment is three fifths, and each service tariff cycles through one, five quarters, and three halves. Training uses quartic coefficient one and graph strength one quarter. Cycle and random-edge tests use those same coefficients; geometric tests use coefficients two and one half; weighted-block tests use three and one. The last graph has within-block connection probability seven tenths, cross-block probability twelve hundredths, and integer edge weights from one to four. Connectivity is not imposed or required for strong convexity. Primitive resource shocks are bounded rational numbers and centered with the actual discounted occupation weights.

The learned potential applies softplus nonlinearities at three scales and seven shifts to each of three local transforms: identity, one smoothing operation, and two smoothing operations. Its 63 readout coefficients are learned by regularized least squares. Both objectives receive the same 72 scalar oracle values per training seed. The derivative-weighted objective adds 24 central-difference equations computed from those same values. The fixed ridge coefficient is one hundred-thousandth. Ten training seeds and two problem-generation seeds per family are disclosed in code. No fastest seed, favorable instance, or failed transfer is selected for reporting.

Nonlinear Newton steps use the exact quartic Hessian. The participation dual is optimized with non-negative multipliers; the dual's floating-point exit status is retained only as a diagnostic. Newton-PCG precomputes one shifted baseline Hessian factorization per graph-tree instance and reuses it across changing Newton systems and all 15 repeated solves. Thus the direct method's changing factorization cost is compared with an actual cached alternative, rather than treating repeated factorization as the only classical baseline. Setup is recorded separately; feature construction, acceptance repair, multiplier proposal, and fraction-based audit are included in the online candidate timings. Training time is separately retained and is not amortized

into a claimed break-even point.

The exact repair first clips tiers to the compact box and rounds downward to a grid with denominator one hundred million. Because tariff entries are positive, multiplying all nonnegative tiers by the smallest subtree-budget ratio preserves the box and makes every payment inequality feasible. A second downward rounding preserves feasibility. The repair is deliberately simple, not claimed to be the closest feasible projection; its economic distortion is visible in the raw rewards. Consequently the static gate compares the repaired proposal with the exactly optimized static protocol before deployment.

The replay checker reconstructs the entire polynomial gradient from the recorded primitives, including both appearances of every temporal edge and both endpoints of every spatial edge. It reconstructs each ancestor multiplier contribution independently. No numerical tolerance is used for acceptance, multiplier signs, objective equality, or equality of the stored rational certificate. Structural checks additionally compare 120 edge-criterion cases with normalized linear programs, verify the tree summation identity with fractions, check 12 absolute-friction cases, and replay the zero-friction promise example with gap one thirty-second. These finite checks exercise the theorem implementation; the proofs do not rest on those tests.

A preliminary unanchored-readout run prompted the explicit reporting of severe-shift failures and the exact static gate. The final fitting architecture, sampling seeds, and regularization were not tuned to remove those failures. The final dataset differs from the preliminary calculation in replacing unchecked floating oracle values with feasible rationally certified values and explicit endpoint error bounds. Both value-only and derivative-weighted objectives receive this same change. All final negative cases remain disclosed; statistical inference concerns ten paired training-seed means, not a fictitious sample of 160 independent fitted networks.

EC.9.1 Cached-factor re-audit of the quadratic diagnostic

The unchanged R7 graph instances and the prespecified first learned-gradient training seed are replayed with cached sparse LU, cached dense Cholesky, Jacobi-preconditioned conjugate gradients, the analytic polynomial filter, and a fresh sparse solve. For each method, the two right-hand sides used to calibrate acceptance are solved together where supported. Setup is excluded from online time but reported separately. Each timing includes the same exact rational certificate; all 72 final decisions and all repeated observations are retained. Table 8 replaces any interpretation of the old per-call factorization comparison as a fair amortized speedup result. The new comparison is still not at matched certified accuracy, and no universal learned advantage is asserted.

EC.9.2 Operator shift diagnostics

Table 9 reports the extremal eigenvalues of the discounted-weight-normalized Hessian at the static tier, as well as maximum weighted spatial degree. The training-sized spectrum is not used as a purported enclosure for shifted graphs. These floating diagnostics are distinct from the exact, dimension-independent maintenance-curvature lower bound used in each residual certificate.

Table 8: R10 re-audit of the unchanged quadratic portfolio task. End-to-end time includes multiplier calibration, rational repair, and exact audit. Setup is paid once per graph; four graph-specific medians are averaged, each using 15 repetitions after one warmup. The learned coefficients use the prespecified first training seed. Accuracy is not matched.

Services	Method	Setup (ms)	End-to-end (ms)	Max. bound/service
64	Sparse, fresh	0.00	1.10	5.61e-12
64	Sparse, cached	0.21	0.96	5.61e-12
64	Dense, cached	0.14	0.98	5.61e-12
64	Jacobi-PCG	0.10	3.00	6.03e-12
64	Analytic filter	0.00	1.13	3.49e-07
64	Learned gradient	0.00	1.11	1.31e-07
256	Sparse, fresh	0.00	2.46	7.25e-12
256	Sparse, cached	0.61	1.90	7.25e-12
256	Dense, cached	0.72	2.06	7.25e-12
256	Jacobi-PCG	0.10	4.15	7.57e-12
256	Analytic filter	0.00	2.04	3.82e-07
256	Learned gradient	0.00	2.02	5.79e-07
1024	Sparse, fresh	0.00	12.41	7.8e-12
1024	Sparse, cached	7.35	5.33	7.8e-12
1024	Dense, cached	12.92	14.13	7.8e-12
1024	Jacobi-PCG	0.18	7.86	7.64e-12
1024	Analytic filter	0.00	5.49	3.67e-07
1024	Learned gradient	0.00	5.47	3.42e-07

Table 9: Measured spectrum of the discounted-weight-normalized Hessian at the static tier and maximum weighted spatial degree. These floating diagnostics expose distribution shift; exact certificates use the proved lower curvature bound instead.

Instance	Coordinates	Minimum	Maximum	Weighted degree
training-template	28	6.26	11.71	2
cycle-4-3-33001	28	6.26	11.71	2
cycle-4-3-33002	28	6.26	11.71	2
erdos-8-3-33001	56	6.26	12.92	5
erdos-8-3-33002	56	6.26	13.35	6
geo-8-4-33001	120	9.17	19.63	5
geo-8-4-33002	120	9.17	19.41	5
block-32-5-33001	992	12.12	112.72	43
block-32-5-33002	992	12.12	128.05	46

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